Classification of States

Definition (Accessibility).
State j is **accessible** from state i, denoted by $i \to j$, if

$$p_{ij}^{(n)} > 0 \quad \text{for some } n \geq 0.$$

Remark.

- Accessibility is not a one-step concept: p_{ij} may be zero while $p_{ij}^{(n)} > 0$ for some $n > 1$.
- It is not symmetric: $i \to j$ does not imply $j \to i$.
- Since $p_{ii}^{(0)} = 1$, every state is accessible from itself.

Definition (Communication).
States i and j **communicate**, denoted by $i \longleftrightarrow j$, if

$$i \to j \qquad \text{and} \qquad j \to i.$$

Note. Communication is an **equivalence** relation.

Definition (Communication Class).
A **communication class** is a maximal set $\mathscr{C} \subseteq S$ whose states communicate with one another.

Note. Communication classes form a partition of the state space S.

Definition (Irreducible chains).
A Markov chain is **irreducible** if it has exactly one communication class. Otherwise, it is **reducible**.

Definition (Closed Class).
A communication class $\mathscr{C}$ is **closed** if the chain cannot leave it:

$$p_{ij} = 0 \qquad \text{for every } i \in \mathscr{C} \text{ and } j \notin \mathscr{C}.$$

Remark.

- Once $X_n \in \mathscr{C}$ for any n, all future states X_{n+m} remain in $\mathscr{C}$ with probability 1, any $m \geq 0$.
- An **absorbing state** i satisfies $p_{ii} = 1$. Hence $\{i\}$ is a closed class.
- A closed class may contain several states and the chain may continue moving inside it.

Theorem.
Every finite-state Markov chain has at least one closed communicating class.